Connor Floyd
UBC Engineering Physics

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Seeking Internships: May 2027 – August 2028

Education

4th Year Engineering Physics **GPA 89.5%**
University of British Columbia, Vancouver, Canada Expected graduation: May 2029
UBC Trek Excellence Scholarship (2024, 2026) Dean's list (2024, 2025); Dean's Scholar (2026)

Skill Sets

Design: **SolidWorks** (large assemblies, parametric modelling, drawings, FEA), **PCB Design** (KiCad)
Manufacturing: **CNC Mill** (Tormach), **Lathe**, Manual Mill, Waterjet, Laser Cutter, **FDM/Resin 3D Printing**, MIG Welding
Test & Instrumentation: **Data Acquisition**, LCR & Micro-Ohm Meters, **Thermocouple Instrumentation**, Test Plans & Reports
Programming & Embedded: **Python** (Pandas, NumPy), C++, MATLAB, **Git**, Linux, ESP32, Raspberry Pi

Technical Experience

Test Engineering Co-op May 2026-August 2026
Corvus Energy

- Planned and executed **electrical characterization tests** on marine battery systems, measuring power-path, contactor and fuse resistance and developing a configuration-based capacitance model
- **Validated safety fault response** by simulating cell over-temperature events and measuring system reaction time at the customer terminals with a **DAQ system**
- Developed a **Python thermal-cycle fatigue tool**, enabling comparison of customer systems against accelerated life tests to **predict system failures** and **validate reliability estimates**.
- Built a **Python tool to process operational data** from customer vessels and **calculate system availability metrics**
- **Built hardware test setups and equipment under test**, including battery modules with internal thermocouples, EMC testing jigs, system control computers, and acoustic barriers
- Built an **insulated thermal enclosure** with chiller integration, **reducing accelerated life cycle time by ~33%**
- Supported **product certification testing** by documenting procedures, deviations, results, and calibration data

Mechanical Engineering Co-op January 2025 - April 2025
Corvus Energy – Prototyping Team

- Designed marine battery system parts and assemblies in **SolidWorks**, using parametric models and FEA validation
- **Manufactured and assembled prototype parts** and battery subassemblies using **CNC machines** and shop tools
- Executed **battery system tests** on thermal runaway, vibrations, adhesives, and environmental sealing
- Wrote **formal documentation** including design documents, risk analyses, test plans, test journals, and reports
- Owned work from ideation to finalized prototypes, **completing 15+ projects** in 4 months with minimal supervision

Engineering Design Team Captain September 2023 - Present
UBC ThunderBikes

- Designed and fabricated parts for an electric race motorcycle, including a **tubular subframe**, battery casings, and high-voltage electronics mounts for a **400A, 110V system**
- Led a team of **55+ members**, managing project timelines, budgeting, and cross-team integration
- Provided technical guidance during design and manufacturing, focusing on the **battery pack** and **high-voltage safety**
- Produced CAD models and drawings in **SolidWorks**, emphasizing ease of manufacturing and assembly
- Performed **finite element analyses** and hand calculations to ensure the safety and robustness of designs

Autonomous Pet Rescue Robot May 2025 - August 2025
ENPH 253 Robotics Competition

- **Led mechanical design** for a 4-person team developing an autonomous robot to locate and transport stuffed animals
- Rapidly tested and iterated during the **6-week competition**, completing 4+ iterations per subsystem
- Created a full **SolidWorks** assembly of the robot, designing subsystems including the chassis, drivetrain, robotic arms, and pickup mechanism
- Designed dual arm pickup system, utilizing a central driveshaft to maintain arm synchronization
- Fabricated components using laser cutting, **machining**, **3D printing**, and hand-tools